DS	**Session**	Tree 1	**Question Information**	Level 1 Challenge 63

Question description

You are given an n×n grid representing the map of a forest. Each square is either empty or contains a tree. The upper-left square has coordinates (1,1), and the lower-right square has coordinates (n,n).

Your task is to process q queries of the form: how many trees are inside a given rectangle in the forest?

Constraints

- 1≤n≤1000
- 1≤q≤2⋅10^5
- 1≤y1≤y2≤n
- 1≤x1≤x2≤n

Input

The first input line has two integers n and q: the size of the forest and the number of queries.
Then, there are n lines describing the forest. Each line has n characters: . is an empty square and * is a tree.
Finally, there are q lines describing the queries. Each line has four integers y1, x1, y2, x2 corresponding to the corners of a rectangle.

Output
Print the number of trees inside each rectangle.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4 3
.*..
*.**
**..
****
2 2 3 4
3 1 3 1
1 1 2 2
```

EXPECTED OUTPUT

```
3
1
2
```

Test Case 2

INPUT (STDIN)

```
4 3
.*.*
*.**
**.*
***.
1 2 3 4
3 2 3 4
1 4 2 3
```

EXPECTED OUTPUT

```
6
2
0
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
for(i=1;i<=n;i++)
```

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
8	310	28